

When Graph Structure Becomes a Liability: A Comprehensive Benchmark of GNN Architectures for Temporal Fraud Detection on Bitcoin Transaction Networks

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<https://github.com/Saket-Maganti/gnn-fraud>

Abstract. Graph Neural Networks (GNNs) are widely assumed to outperform feature-only baselines on financial fraud detection because transaction networks contain structural signals that individual node features cannot capture on their own. This paper tests that assumption under a stricter evaluation protocol.

We benchmark four architectures (MLP, GCN, GraphSAGE, and GAT) across three class-imbalance strategies on the Elliptic Bitcoin Dataset (203,769 nodes, 234,355 edges, 49 temporal steps). The protocol uses inductive mini-batch evaluation, three-seed statistical validation, and a set of targeted analyses covering temporal drift, calibration, feature attribution, business cost, and hardware constraints for temporal GNNs. The central result is that a plain three-layer MLP reaches $F1 = 0.744 \pm 0.005$ and outperforms every GNN. GraphSAGE, the strongest graph model in our benchmark, reaches only $F1 = 0.697 \pm 0.009$.

The explanation is temporal distribution shift. The illicit rate falls from 11.6 % during training to 0.3 % in the hardest test steps, and every model collapses after step 42. Random Forest analysis shows that 82.4 % of discriminative power comes from local transaction features, while graph structure becomes unreliable once the fraud topology changes. Temperature scaling leaves F1 unchanged, which rules out simple calibration fixes. Business-cost analysis shows that the highest-F1 model is not always the cheapest one to operate under asymmetric false-negative and false-positive penalties.

Keywords: Graph neural networks · financial fraud detection · temporal distribution shift · Elliptic Bitcoin Dataset · concept drift

1 Introduction

Financial fraud detection on transaction networks has become a standard use case for Graph Neural Networks. The intuition is straightforward. Fraud rings

share infrastructure, layering schemes create transaction chains, and mixing activity leaves structural motifs that are difficult to recover from isolated node features alone. On that logic, a graph model should outperform a feature-only baseline.

The Elliptic Bitcoin Dataset [8] has reinforced that view. Published work reports strong results for GCN, GraphSAGE, and EvolveGCN [1, 6–8]. Yet two details in that literature make the headline comparison less decisive than it first appears. First, prior studies rely on transductive evaluation, where the test graph is available during training. Second, they report aggregate test metrics across the full test window rather than showing how performance changes from one time step to the next.

This paper revisits the benchmark under inductive evaluation and a per-step temporal analysis. The result is not a small correction. A three-layer MLP beats every GNN in aggregate F1, and all models collapse once the test period enters a much lower-fraud regime.

Contributions

- A three-seed benchmark covering four architectures and three imbalance strategies under inductive evaluation.
- A per-time-step analysis showing that the aggregate test score hides a sharp collapse after step 42.
- Feature-attribution and community-structure analysis explaining why the MLP stays competitive and why graph structure becomes fragile.
- Calibration and hardware analyses that rule out simple fixes and clarify the practical requirements of temporal GNN deployment.

2 Related work

Weber et al. [8] introduced the Elliptic benchmark and a GCN baseline. Alarab et al. [1] improved results with extra feature engineering. Pareja et al. [7] extended the setting to temporal graph learning with EvolveGCN. Lo et al. [6] reported a strong GraphSAGE result with augmentation. All of these studies support the idea that graph structure helps, but they do so under transductive evaluation and aggregate reporting.

Class imbalance is a second thread in this literature. SMOTE [2], GraphSMOTE [9], and focal loss [5] all try to increase minority-class sensitivity. Calibration methods such as temperature scaling [4] adjust probability outputs after training, while concept-drift work [3] studies the moving-target problem that fraud detection systems face in practice.

3 Dataset and experimental setup

3.1 Elliptic Bitcoin Dataset

The dataset contains 203,769 transactions, 234,355 edges, and 165 node features. Of the 46,564 labeled nodes, 4,545 are illicit and 42,019 are licit. The standard temporal split trains on steps 1–34 and evaluates on steps 35–49.

Table 1. Elliptic dataset summary.

Property	Value
Nodes	203,769
Edges (undirected)	234,355
Features	165 (94 local + 71 aggregate)
Labeled nodes	46,564
Illicit nodes	4,545
Licit nodes	42,019
Training split	Steps 1–34; 11.6 % illicit
Test split	Steps 35–49; 6.5 % illicit overall

Figure 1 shows the temporal and class-distribution context that the table alone cannot capture. The test window is not just later in time; it also operates under a different fraud-rate regime.

3.2 Models, strategies, and evaluation

We compare an MLP, GCN, GraphSAGE, GAT, and a CPU-constrained EvolveGCN ablation. The MLP acts as the feature-only control. The GNNs use the same node features plus learned neighbourhood aggregation. We evaluate three imbalance strategies: Baseline cross-entropy, Weighted cross-entropy, and graph augmentation. The main metric is illicit-class F1, reported as mean $\pm$ std across three seeds.

Evaluation is inductive. We use random-node mini-batches so that the model does not access future test neighbourhoods during training. This is stricter than the transductive setup used in prior Elliptic work.

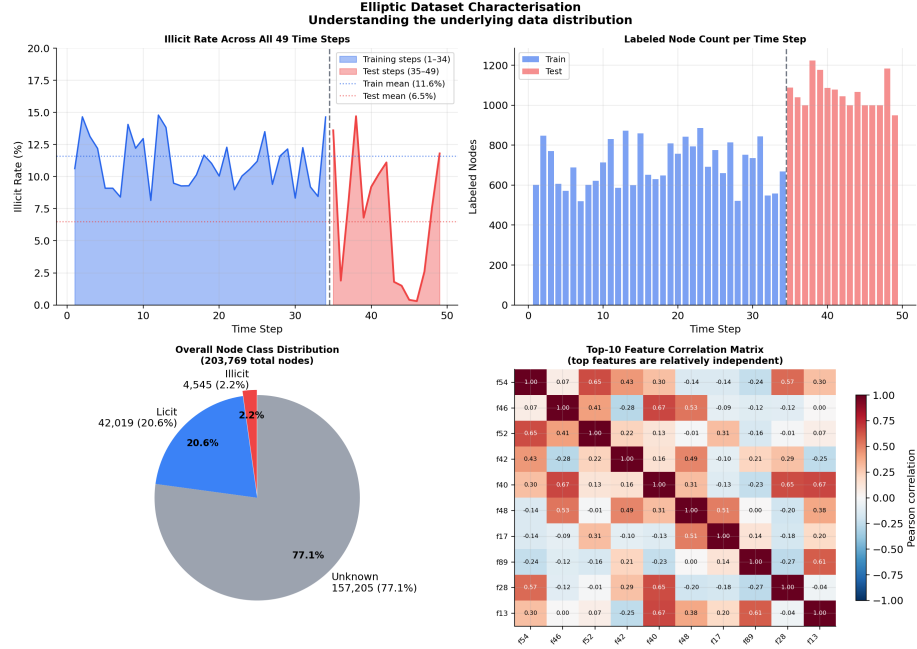


Fig. 1. Dataset characterisation. The temporal split coincides with a marked change in illicit-rate behaviour, and the strongest features remain informative without collapsing into a single redundant block.

4 Results

4.1 Main benchmark

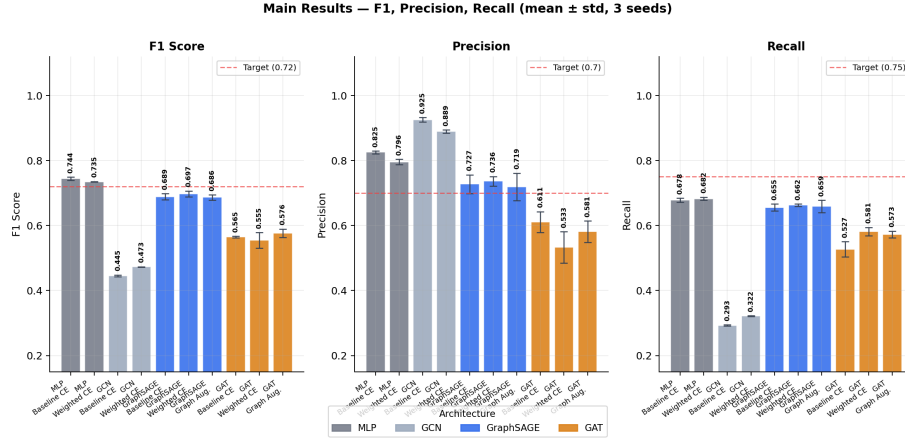
Figure 2 and Table 2 summarise the main comparison. The MLP is the best model in aggregate F1. GraphSAGE is the strongest GNN, while GCN shows a large precision-recall imbalance.

4.2 Strategy sensitivity and robustness

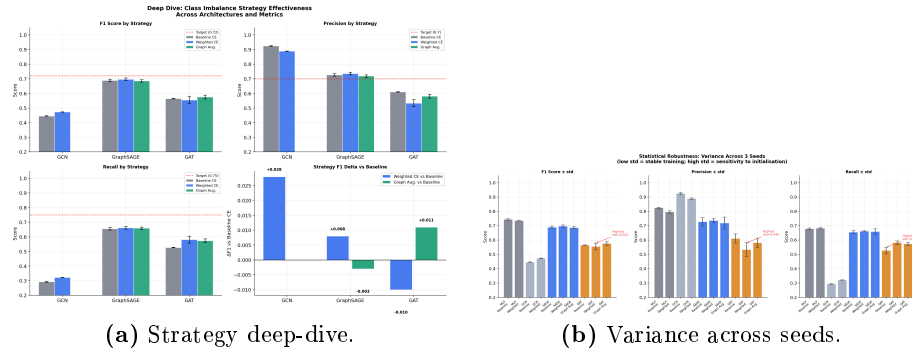
Figure 3 reorganises the benchmark around the imbalance strategy and the seed variance. The main point is that architecture matters more than strategy in the final ranking, although the strategy changes the precision-recall trade-off.

4.3 Temporal collapse

The central empirical result is temporal collapse. Figure 4 shows that every model degrades sharply after step 42, which is exactly where the test period moves into a lower-fraud regime.

**Fig. 2.** Main benchmark across all configurations.**Table 2.** Main benchmark results (mean $\pm$ std, 3 seeds).

Model	Strategy	F1	Prec.	Rec.	AUC
MLP	Baseline	0.744 $\pm$ 0.005	0.825 $\pm$ 0.004	0.678 $\pm$ 0.006	0.891 $\pm$ 0.001
	Weighted	0.735 $\pm$ 0.001	0.796 $\pm$ 0.008	0.682 $\pm$ 0.005	0.893 $\pm$ 0.006
GCN	Baseline	0.445 $\pm$ 0.003	0.925 $\pm$ 0.007	0.293 $\pm$ 0.002	0.868 $\pm$ 0.009
	Weighted	0.473 $\pm$ 0.001	0.889 $\pm$ 0.005	0.322 $\pm$ 0.001	0.872 $\pm$ 0.004
GraphSAGE	Baseline	0.689 $\pm$ 0.009	0.727 $\pm$ 0.029	0.655 $\pm$ 0.011	0.867 $\pm$ 0.006
	Weighted	0.697 $\pm$ 0.009	0.736 $\pm$ 0.015	0.662 $\pm$ 0.004	0.868 $\pm$ 0.003
	Graph Aug.	0.686 $\pm$ 0.008	0.719 $\pm$ 0.042	0.659 $\pm$ 0.019	0.861 $\pm$ 0.007
GAT	Baseline	0.565 $\pm$ 0.002	0.611 $\pm$ 0.032	0.527 $\pm$ 0.023	0.863 $\pm$ 0.007
	Weighted	0.555 $\pm$ 0.024	0.533 $\pm$ 0.048	0.581 $\pm$ 0.013	0.855 $\pm$ 0.004
	Graph Aug.	0.576 $\pm$ 0.013	0.581 $\pm$ 0.033	0.573 $\pm$ 0.010	0.842 $\pm$ 0.002

**Fig. 3.** Strategy sensitivity and statistical robustness.

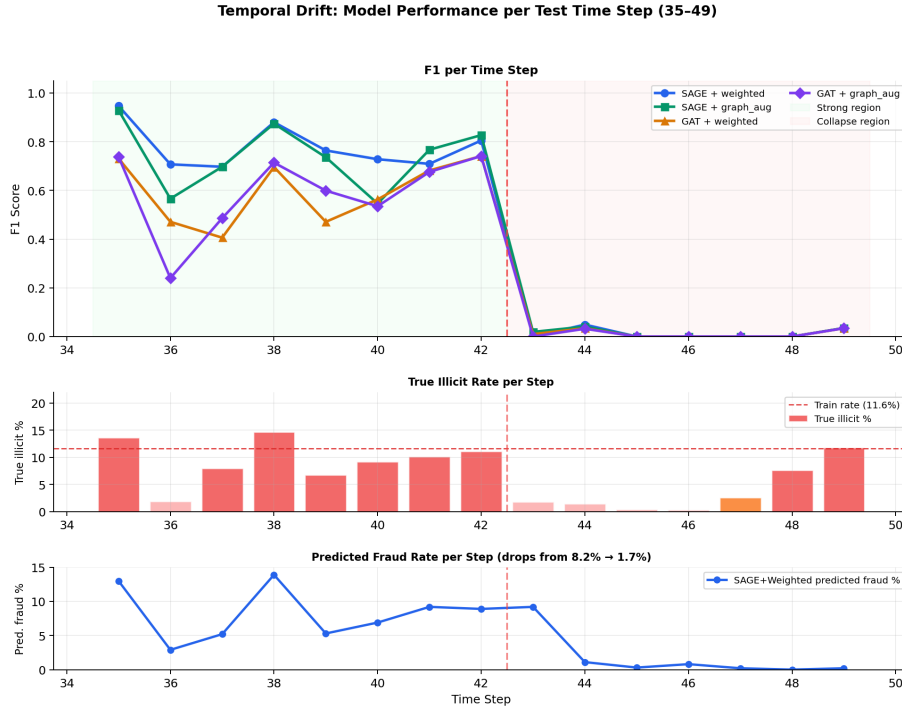


Fig. 4. Per-step F1 and illicit-rate shift across the test window.

This is a recall failure rather than a generic loss of discrimination. The models become increasingly conservative and stop predicting fraud in the late steps. Temperature scaling does not repair the collapse.

4.4 Why the MLP stays ahead

Figure 5 shows two parts of the explanation. Local transaction features dominate the Random Forest importance ranking, accounting for 82.4 % of total importance, while fraud community structure is clearly present in the graph. Figure 6 connects those two observations: graph structure can help briefly, but the advantage flips after the regime change.

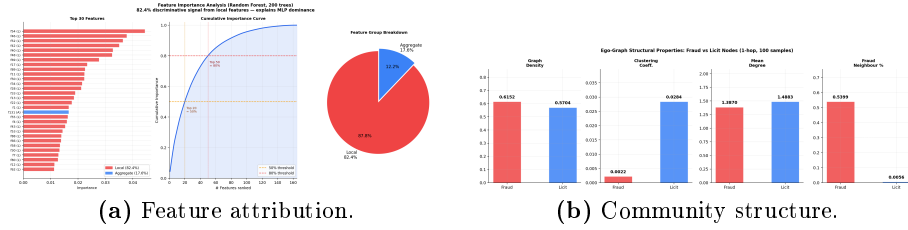


Fig. 5. Local features dominate the stable signal, while community structure remains informative but temporally fragile.

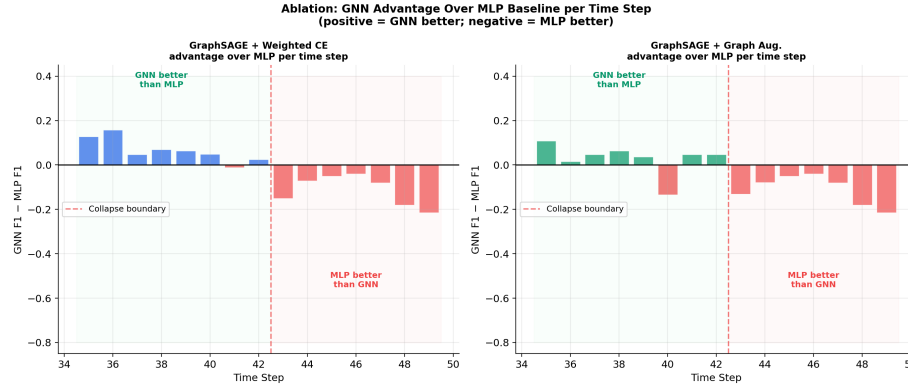


Fig. 6. Per-step GraphSAGE advantage over the MLP baseline. The sign reversal after step 42 shows when graph structure turns from a short-term asset into a liability.

5 Additional analyses

Temperature scaling produces fitted temperatures in the mild-overconfidence range, but it leaves F1 unchanged, which indicates that the failure is not a simple calibration issue. The EvolveGCN ablation reaches $F1 = 0.29$ on CPU, far below the published GPU result of 0.77, which shows how strongly temporal architectures depend on full snapshots and adequate hardware. Business-cost analysis under asymmetric penalties further shows that the highest-F1 model is not always the most cost-effective one to deploy.

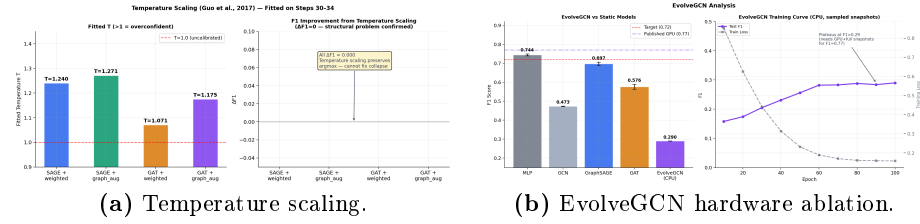


Fig. 7. Post-hoc calibration does not recover performance, while temporal GNN deployment remains hardware-sensitive.

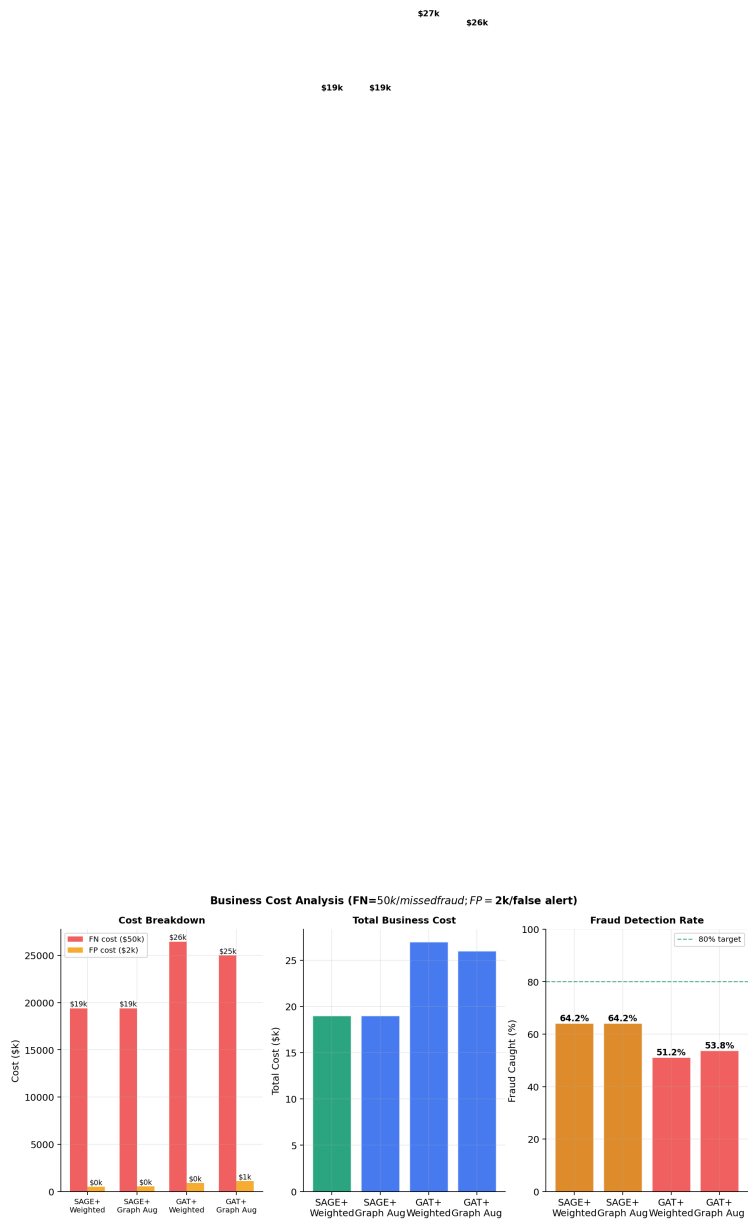


Fig. 8. Business-cost analysis under asymmetric false-negative and false-positive penalties.

6 Discussion and conclusion

Three practical conclusions follow from the benchmark. First, periodic retraining is necessary when the fraud rate moves this quickly. A static model trained under one base rate cannot be expected to stay calibrated under another. Second, graph structure is a conditional asset rather than a universal one. It can help in stable periods and become harmful once the topology drifts. Third, fraud-detection models should be selected against operational cost functions, not against F1 alone.

The study has clear limitations. It is based on one dataset, and the inductive pipeline uses RandomNodeLoader rather than the full neighbour-sampling setup that would be preferable on hardware with broader support. Even with those limitations, the main result is hard to ignore: under stricter evaluation, the feature-only baseline is stronger and more robust than the graph models tested here.

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